

TEAM XNNOV473RS!

THEME FINTECH



PMS - PENSION MANAGEMENT SYSTEM

Financial Intelligence for Informal sector



PROBLEM STATEMENT

THE ISSUE

Daily wage earners in the informal sector face destitute old age because irregular cash flows, lack of digital footprints, and rigid banking systems make consistent retirement planning impossible.

WHY THIS HAPPENS?



Earnings are erratic; fixed monthly SIPs fail.



Formal banks are distant and intimidating; visiting a branch means losing a day's wage.



Workers fear "locking" money away for 30 years because they might need it tomorrow for an emergency.

1

Flexibility Gap (Product):

- Current pension schemes demand fixed monthly amounts. No system allows "micro-saving" (e.g., ₹20) based on daily surplus.



2

Data Gap (Verification):

- Cash earnings leave no digital trail. Without payslips, these workers have no "Credit Score" and are invisible to lenders.



KEY
GAPS

4

Trust Gap (Behavioral):

- Low digital literacy and fear of scams make workers hoard cash instead of trusting digital wallets or apps.



3

Resilience Gap (Safety):

- One medical emergency often forces them to withdraw all savings, killing their compound interest growth instantly.



EXCLUSION BY DESIGN



Current financial instruments are designed for the salaried class. They identify "failure to pay" as indiscipline, rather than understanding the reality of irregular income.

PROBLEM : THE INVISIBLE TRAP OF INFORMAL LABOR



Unrecorded Income

Zero Collateral

Unlocalized triggers

Inaccessible monitoring

Today, the informal sector has two unconnected realities: Hard daily labor vs. Zero long-term financial security. No system bridges them.



HARD CASH. ZERO CREDIT.



Earnings are physical, volatile, and temporary. Financial security is digital, stable, and long-term. Currently, they never meet..

SOLUTION ABSTRACT

PMS - PENSION MANAGEMENT SYSTEM Financial Intelligence for Informal sector

Smart, agentic micro-pension ecosystem that converts volatile cash earnings into stable long-term financial security

FEATURES



Income Pattern Sensing

Tracks daily earnings volatility using adaptive financial signal processing.



Context-Aware Financial Decisions

Adjusts savings targets based on income stability, weather, and work patterns.



Predictive Financial Intelligence

Detects risk of savings drop or income disruption before it happens.



Proactive Financial Guidance

Suggests safe savings amounts without affecting daily survival needs.



Privacy & Social Governance

Dual-key approvals + verified income ledger prevents spending leakage.

IMPLEMENTATION EASE

Designed for plug-and-play adoption using existing payment networks and local retail agents - no new infrastructure required

EFFECTIVENESS



Prevents savings drop during income shocks



Converts physical cash → digital pension seamlessly



Reduces financial vulnerability in emergencies

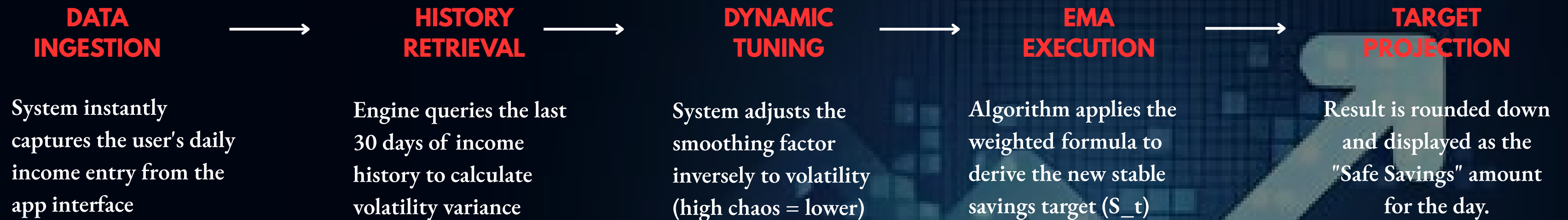


Enables long-term wealth building for informal workers

METHODOLOGY

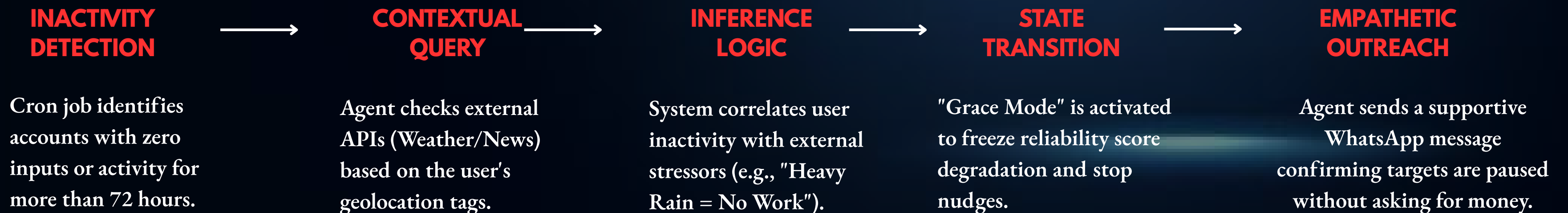
FINANCIAL SIGNAL PROCESSING ENGINE

WORKFLOW: Adaptive Target Calculation



THE GUARDIAN AGENT

WORKFLOW: Context-Aware Automation



METHODOLOGY

THE SOURCE-VERIFIED LEDGER PROTOCOL

WORKFLOW: Payer-Led Income Verification

TRANSACTION TRIGGER

Payer scans the worker's QR code to initiate a digital payment or performance rating

DATA INJECTION

Payer inputs the specific amount paid directly into the verified web interface

IMMUTABLE RECORD

System flags the transaction as source verified and locks the data entry

DATABASE SYNC

Worker's ledger updates with a "Hard Floor" amount that cannot be edited downwards

READONLY DISPLAY

Verified earnings become visible to the user in a locked field, preventing under-reporting

THE DUAL-KEY GOVERNANCE

WORKFLOW: Multi-Sig Withdrawal Request

WITHDRAWAL INITIATION

Worker initiates a withdrawal request for an amount exceeding the Liquid vault limit.

SMART CONTRACT TRIGGER

Funds move to a "Pending" state and a time-locked OTP is generated.

SECONDARY AUTH

System pushes a specialized, time-sensitive OTP to the nominee

CONSENSUS VERIFICATION

Entering the nominee's OTP, creates a valid dual-signature match for the transaction

FUND RELEASE

Logic gate opens and funds are successfully transferred to the linked bank account

METHODOLOGY

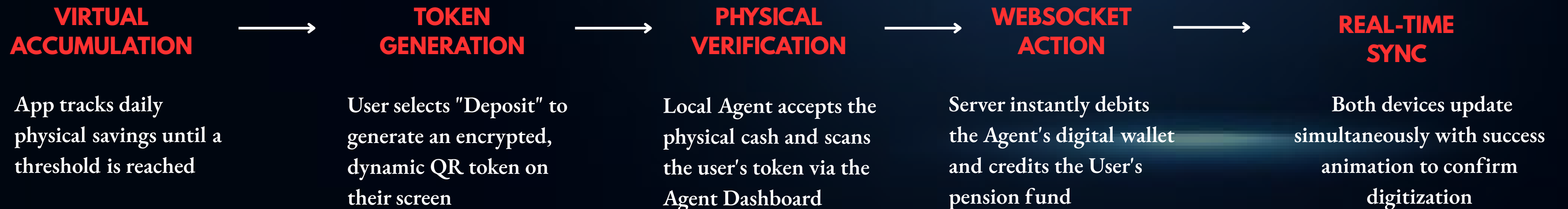
THE EMERGENCY SHIELD

WORKFLOW: Split-Vault Allocation



THE DIGITAL BRIDGE

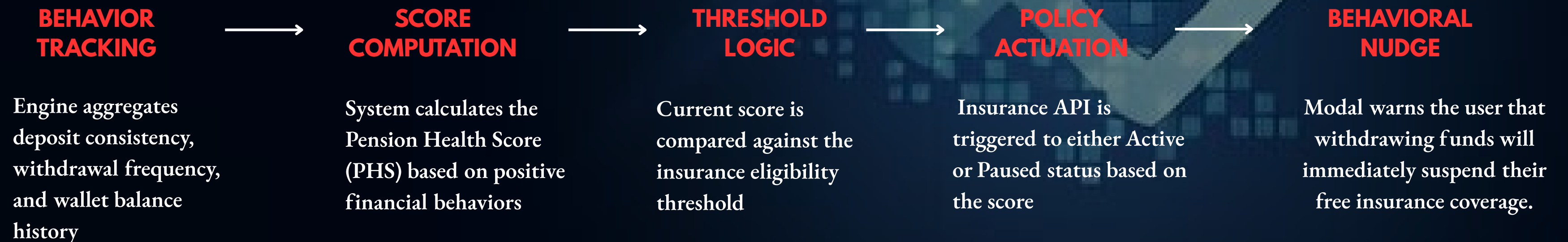
WORKFLOW: Cash-to-Digital Handshake



METHODOLOGY

THE PENSION HEALTH & REWARDS ENGINE

WORKFLOW: Behavior-Based Insurance Toggle

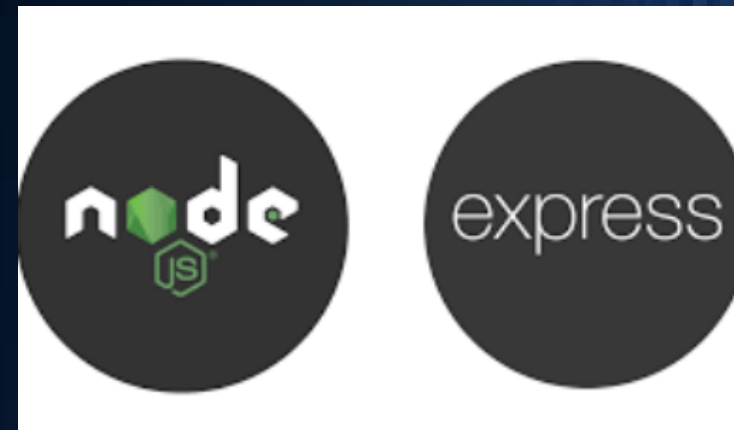


TECH STACK

FRONTEND



BACKEND

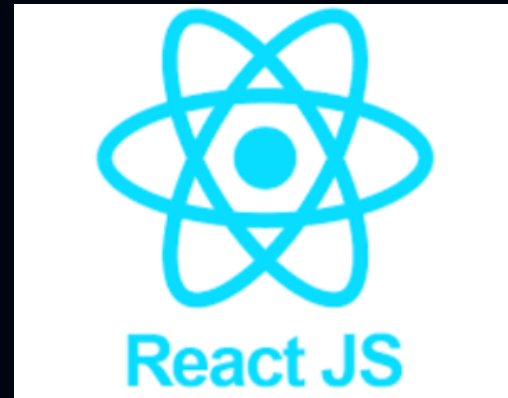


TOOLS & ID'S



TECH STACK - REAL TIME DEPLOYMENT

FRONTEND



ARCHITECTURE



DEPLOYMENT



AUTHENTICATION



CLOUD DEPLOYMENT

Render / Railway / AWS



Real Bank API Integration
(Future scope)

MARKET STUDY & BUSINESS MODEL

EXISTING FINANCE MODALITIES



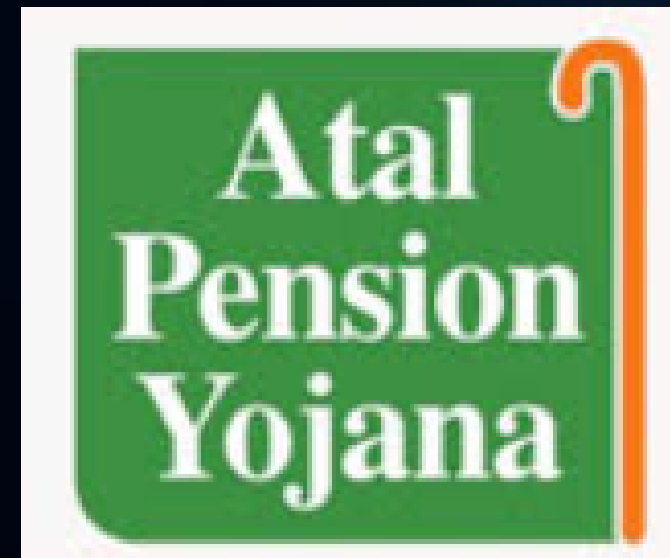
Pension Bee



Pin box Solutions



Jumo



Atal Pension Yojana

Total Addressable Market (TAM): ~85% of India's workforce is informal, creating a massive vacuum for flexible micro-pensions.

Nearly 47+ crore workers operate in informal employment needing adaptive financial tools.

The Cash Barrier: 90% of earnings are physical cash, making digital-only pension apps inaccessible

FUTURE ROADMAP

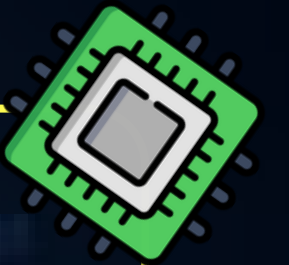
VOICE FINANCIAL ASSISTANT

- Voice alerts for unsafe spending or missed savings
- Voice-based daily income entry for low literacy users
- News-based financial & job-risk alerts using real-time local + economic news feeds



ON-DEVICE INTELLIGENCE

- Convert models to edge-AI for offline income prediction
- Lightweight app for low-end smartphones



CHATBOT

- Conversational financial planning in local languages
- Personalized saving + pension health recommendations



SECURITY LAYER

- End-to-end encrypted financial + identity data
- Multi-party authorization for withdrawals from pension vault



CHALLENGES



- Ensuring zero-loss ledger synchronization in intermittent, low-network environments.
- Validating self-reported cash income without formal salary slips or digital trails.
- Overcoming deep-rooted distrust of digital systems and curbing impulsive spending leakage.

SCOPE



- Autonomous, volatility-adjusted micro-pension management via local agent networks.
- Generative AI voice assistance and alternative credit scoring for emergency micro-loans.



UNIQUE SELLING POINTS

- Dynamic AI targets that adapt to daily income fluctuations (Signal Processing).
- Converts physical cash to digital pension via existing local agent networks.
- "Social Firewall" requires spousal approval to prevent spending leakage on vices.

RESEARCH AND REFERENCES

INDUSTRY OUTLOOK

- ~90% of India's workforce (400M+) is informal; strong government push (PM-SYM, Jan Dhan) supports micro-pension adoption.
- Post-COVID smartphone usage has skyrocketed, creating a perfect window for UPI-based micro-savings over cash.

FINDINGS

- 85% of workers have irregular earnings, making traditional fixed monthly SIPs impossible to sustain.
- 60%+ savings are held in cash/gold due to inflation fears and the need for instant emergency access (no lock-ins).

USER INSIGHTS

- Users distrust "faceless" apps and percentage math, relying instead on local agents and "survival days" logic.
- Fear of losing money to medical emergencies prevents long-term saving; they demand instant withdrawal options.

REFERENCES

- NITI Aayog (Gig Economy Strategy) and RBI Reports (National Financial Inclusion Strategy).
- World Bank (Behavioral Insights on Micro-savings)
- ILO (Social Protection Gaps in Informal Sector).

EXPERT INSIGHTS

- Traditional banks cannot profitably service tiny deposits (₹50); a low-cost, automated tech layer is essential.
- "Nudges" (Streaks) and spousal oversight (Dual-Key) drive habit formation better than financial education.

TECH/MARKET VOICE

- Must use Offline-First (SQLite) architecture to ensure reliability in areas with patchy 4G networks.
- Massive gap in serving the "Bottom 90%"; savings data can create Alternative Credit Scores for micro-loans.

TAKE A LOOK AT XNNQV473R5!

ROLE	NAME	STREAM/ DEPT	YEAR	COLLEGE
Team lead - Backend developer	Harinisri Ramesh	Biomedical Engg	UG III	SSNCE
Backend & ML Engineer	Harini V	Biomedical Engg	UG III	SSNCE
Web & Frontend Developer	Malathi G	Computer Science	UG IV	SSNCE